

NAKHONSI THAMMARAT AGROMET, Thailand

WMO: 485540

Lat: 8.359N

Lon: 100.059E

Elev: 3

StdP: 101.29

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.2	(c) 22.9	(d) 20.8	(e) 15.5	(f) 24.7	(g) 21.4	(h) 16.1	(i) 25.3	(j) 7.7	(k) 28.3	(l) 6.3	(m) 27.3	(n) 0.0	(o) N/A	(p) 0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 5	(b) 8.1	(c) 35.2	(d) 27.4	(e) 34.5	(f) 27.3	(g) 33.9	(h) 27.2	(i) 28.2	(j) 33.2	(k) 28.0	(l) 32.8	(m) 27.7	(n) 32.3	(o) 2.9	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	22.8	30.9	26.7	22.3	30.5	26.5	22.0	30.2	91.1	33.3	89.5	33.1	88.2	32.6	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.3	5.3	4.7	DB	20.7	36.2	1.3	1.0	19.7	36.9	19.0	37.4	18.2	38.0	17.3	38.7	
			WB	19.9	28.9	1.2	0.4	19.1	29.2	18.4	29.4	17.8	29.6	16.9	29.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	26.6	26.9	27.9	29.0	29.3	29.1	28.9	28.9	28.5	27.8	27.0	26.7
	DBStd	1.42	0.98	0.80	0.98	1.33	0.93	0.97	0.99	0.90	0.96	1.04	1.15	1.02
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6591	513	473	554	569	599	574	587	587	554	553	511	516
	CDD18.3	3549	255	240	296	319	341	324	328	329	304	294	261	258
	CDH23.3	35828	2167	2220	3074	3596	3751	3622	3542	3477	3027	2791	2347	2215
	CDH26.7	13241	548	745	1216	1563	1575	1531	1462	1444	1133	916	611	497
Wind	WSAvg	1.1	1.0	0.9	0.8	0.8	0.9	1.3	1.5	1.4	1.2	0.9	0.9	1.2
Precipitation	PrecAvg	1635	73	29	60	75	128	137	146	146	146	216	288	191
	PrecMax	2580	448	172	423	199	202	194	215	231	212	384	704	463
	PrecMin	1049	2	3	2	6	75	47	60	83	90	102	126	52
	PrecStd	285	84	35	80	47	30	33	41	38	30	64	135	107
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.6	32.2	34.4	36.1	35.8	35.4	35.4	35.6	34.6	33.5	32.4	31.8
		MCWB	26.4	26.0	27.5	28.0	27.8	27.3	27.2	26.6	27.0	27.2	26.8	26.8
	2%	DB	30.8	31.5	33.2	35.0	34.7	34.5	34.5	34.6	33.6	32.5	31.5	30.9
		MCWB	26.1	25.9	26.8	27.7	27.7	27.2	27.0	26.8	26.9	27.0	26.9	26.5
	5%	DB	30.2	31.0	32.5	33.9	33.9	33.7	33.6	33.7	32.9	31.8	30.8	30.2
		MCWB	25.8	25.7	26.5	27.3	27.6	27.1	26.8	26.8	26.9	26.9	26.8	26.3
	10%	DB	29.5	30.4	31.7	33.0	33.0	32.9	32.6	32.6	31.9	31.0	30.0	29.4
		MCWB	25.6	25.5	26.3	27.1	27.4	27.0	26.6	26.7	26.7	26.8	26.7	26.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.1	28.1	28.8	28.7	28.5	27.9	28.1	28.0	28.0	27.7	27.6
		MCDB	30.2	30.9	33.1	34.6	33.7	33.7	33.4	33.3	32.5	31.9	30.7	30.2
	2%	WB	26.7	26.6	27.4	28.2	28.2	28.0	27.5	27.6	27.6	27.5	27.4	27.1
		MCDB	29.7	30.4	32.0	33.5	33.1	33.1	32.8	32.6	32.0	31.3	30.3	29.9
	5%	WB	26.3	26.2	26.9	27.7	27.9	27.5	27.2	27.2	27.2	27.2	27.1	26.7
		MCDB	29.1	29.9	31.3	32.7	32.7	32.5	32.3	32.1	31.7	30.8	29.9	29.3
	10%	WB	25.9	25.8	26.5	27.3	27.6	27.1	26.8	26.8	26.8	26.9	26.7	26.3
		MCDB	28.6	29.4	30.6	31.8	32.1	31.8	31.6	31.5	31.2	30.4	29.3	28.6
Mean Daily Temperature Range	5% DB	MDBR	6.0	7.8	8.1	8.1	8.1	8.1	8.1	8.5	7.7	6.8	5.3	5.1
		MCDBR	7.1	8.4	9.3	9.7	9.3	9.2	9.4	9.7	9.0	8.0	7.0	6.6
		MCWBR	2.9	3.4	3.5	3.3	3.3	3.0	3.0	3.2	3.2	3.1	3.0	2.7
	5% WB	MCDBR	6.3	7.8	8.6	9.0	8.7	8.6	8.6	8.8	8.4	7.4	6.3	5.8
		MCWBR	3.0	3.5	3.5	3.4	3.4	3.3	3.1	3.3	3.3	3.3	3.0	2.8
Clear-Sky Solar Irradiance	taub		0.495	0.499	0.507	0.478	0.456	0.453	0.455	0.449	0.442	0.454	0.476	0.486
	taud		2.168	2.148	2.163	2.252	2.352	2.386	2.368	2.373	2.386	2.358	2.274	2.215
	Ebn at Noon		820	830	827	839	841	832	834	852	870	860	832	820
	Edh at Noon		151	158	158	142	126	120	123	124	124	127	135	142
All-Sky Solar Radiation	RadAvg		4.62	5.59	5.80	5.77	5.17	4.96	4.90	5.04	4.93	4.43	3.94	3.93
	RadStd		0.43	0.37	0.49	0.35	0.29	0.26	0.29	0.23	0.20	0.36	0.38	0.42

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	+0.21	+0.25	N/A	+0.31	+0.21	+0.24	N/A	N/A	+74	+75	

Nomenclature: See separate page